Business Plan for Decision Tree Classification Assignment

FreshBasket Subscription Grocery Delivery

Topic: Customer Churn Prediction and Retention Decision Support

Assignment Context	This business plan is linked to the Excel dataset named freshbasket_customer_churn_dataset.xlsx. Students should use the dataset to build a Decision Tree classification model and explain the model results from a business perspective.

1. Business Overview

FreshBasket is a subscription-based grocery delivery service operating in the Greater Toronto Area. The company offers weekly and monthly grocery delivery plans for busy professionals, families, and customers who prefer convenient online ordering. Customers can choose different subscription types, delivery windows, and payment methods. The company earns revenue through recurring subscription fees and regular grocery orders.

Because FreshBasket depends on repeat customers, customer retention is very important. When customers cancel or stop renewing their subscription, the business loses recurring revenue and must spend more money to acquire new customers.

2. Business Problem

The main business problem is customer churn. FreshBasket wants to identify customers who are at risk of leaving the service before they cancel. If the company can predict churn risk early, it can take action such as offering customer support, improving delivery reliability, sending a personalized discount, or recommending a better subscription plan.

3. Machine Learning Decision

Machine learning can help FreshBasket decide which customers should receive retention attention. A Decision Tree classification model can learn patterns from customer behaviour and predict whether a customer is likely to churn.

Business Question	Machine Learning Decision
Which customers are likely to cancel or not renew?	Predict Churned = Yes or Churned = No.
Which factors explain customer churn risk?	Use feature importance from the Decision Tree model.

Where should the business focus retention actions?	Prioritize high-risk customers and improve the strongest churn-related factors.

4. Dataset Description

The dataset is synthetic and created for educational use. It represents customers of FreshBasket and includes demographic, subscription, spending, service-quality, and engagement variables. The dataset also includes a few missing values and duplicate rows so students can practice data inspection and cleaning.

Variable	Role	Type	Business Meaning
Churned	Target variable	Categorical	Whether the customer cancelled or did not renew.
Subscription_Type	Input feature	Categorical	Plan selected by the customer: Basic, Family, or Premium.
Monthly_Spend	Input feature	Numerical	How much the customer spends each month.
Orders_Last_3_Months	Input feature	Numerical	Recent purchasing activity.
Delivery_Issues_Last_3_Months	Input feature	Numerical	Recent delivery problems.
Customer_Service_Tickets	Input feature	Numerical	Number of support requests.
App_Engagement_Score	Input feature	Numerical	How actively the customer uses the app.
Satisfaction_Score	Input feature	Numerical	Customer satisfaction rating from 1 to 5.

5. Target Variable and Input Features

Target variable:

Churned — This indicates whether the customer left the subscription service. The values are Yes and No.

Input features:

- Age_Group
- Region
- Subscription_Type
- Payment_Method
- Delivery_Window
- Membership_Length_Months
- Monthly_Spend
- Average_Order_Value
- Orders_Last_3_Months
- Delivery_Issues_Last_3_Months
- Customer_Service_Tickets
- Discount_Used
- App_Engagement_Score
- Satisfaction_Score

6. Why This Prediction Is Useful

- The business can identify high-risk customers before they leave.
- The company can reduce revenue loss by targeting retention actions more effectively.
- Managers can understand which factors, such as delivery problems or low satisfaction, are most connected to churn.
- The business can improve operations by focusing on the customer experience issues that matter most.
- Marketing and customer support teams can use model results to prioritize limited time and resources.

7. Business Interpretation Guide for Students

After training the Decision Tree model, students should not only report technical metrics. They should explain what the metrics mean for FreshBasket.

Model Result	Business Interpretation
Accuracy	Overall percentage of correct predictions, but it may hide problems if churned customers are a smaller group.
Precision	Among customers predicted to churn, how many actually churned. This matters when retention offers are costly.
Recall	Among customers who actually churned, how many the model detected. This is important if missing at-risk customers is expensive.
F1-score	A balanced metric when both false positives and false negatives matter.

Training vs. testing accuracy	A large gap may show overfitting, meaning the tree learned training data too closely and may not generalize well.
Feature importance	Shows which customer factors most influenced the tree's predictions and can guide business action.

8. False Positives and False Negatives

Error Type	Meaning in This Dataset	Business Consequence
False Positive	The model predicts Churned = Yes, but the customer would not actually churn.	The business may spend unnecessary retention resources, such as discounts or support calls.
False Negative	The model predicts Churned = No, but the customer actually churns.	The business may fail to help an at-risk customer and lose recurring revenue.

9. Recommended Business Actions

- Contact high-risk customers with a service recovery message or support follow-up.
- Offer a targeted discount only when the cost is justified by churn risk.
- Investigate delivery issues in regions or delivery windows connected to higher churn.
- Improve app engagement through reminders, personalized recommendations, or easier ordering.
- Use feature importance to identify operational changes, not only marketing actions.

10. Limitation and Responsible AI Reflection

This dataset is synthetic and simplified. It may not fully represent real customer behaviour, seasonal changes, income differences, local service conditions, or unexpected events. A Decision Tree can also overfit if it becomes too deep, so students should compare training and testing accuracy.

Human judgment should still be used because model predictions are decision-support tools, not final business decisions. Managers should review customer context, avoid unfair treatment, protect privacy, and make sure retention actions are helpful rather than intrusive.

11. Suggested README Summary

README Section	What Students Should Write
Business problem	FreshBasket wants to predict customer churn and

	improve retention decisions.
Dataset used	FreshBasket customer churn dataset with customer, service, spending, and engagement features.
Model used	Decision Tree Classifier.
Main results	Confusion matrix, accuracy, precision, recall, F1-score, training accuracy, and testing accuracy.
Key insights	Important features that help explain churn risk and possible business actions.
Limitation	Synthetic data and possible bias or overfitting.